0.1 Finite Volume Method

The finite volume method (FVM) is a discretization method widely used in various scenarios that conservation laws are involved. The finite volume is a robust algorithm that can be applied to arbitrary complex geometry. The most important aspect of FVM that benefits this simulation is that the local mass conservation is achieved naturally. To begin with, we consider a general hyperbolic conservation law regarding a scalar variable u as

$$\begin{cases} \dfrac{\partial u(\mathbf{x},t)}{\partial t} + \nabla \cdot \mathbf{f}(u;\mathbf{x},t) = 0, & t > 0, \mathbf{x} \in \Omega, \\ u(\mathbf{x},0) = u_0(\mathbf{x}), & \mathbf{x} \in \Omega, \end{cases} \tag{1}$$

where we assume the domain $\Omega \subset \mathbb{R}^2$. The term $\mathbf{f}$ is a flux transporting scalar variable u. We write the basic semi-discretized form of basic finite volume method as

$$\frac{\partial \overline{u}}{\partial t} + \frac{1}{|E|}\int_{\partial E} \mathbf{F}(\mathbf{x},t) \cdot \boldsymbol{\nu} d\mathbf{x} = 0, \tag{2}$$

where we use the cell-averaged value

$$\overline{u} = \frac{1}{|E|}\int_E u(\mathbf{x},t)\, d\mathbf{x}, \tag{3}$$

and $\mathbf{F}(\mathbf{x},t)$ is the approximated numerical flux. We use the Lax-Friedrichs numerical flux

$$\mathbf{F}(u^+,u^-) = \frac{1}{2}[(\mathbf{f}(u^+) + \mathbf{f}(u^-)) \cdot \boldsymbol{\nu} - \alpha_{LF}(u^+ - u^-)], \tag{4}$$

where u^+ and u^- represent values approximated from element $E^+ \in \mathcal{T}_h$ and $E^- \in \mathcal{T}_h$ respectively, and $\boldsymbol{\nu}$ is the unit vector pointing from E^+ to E^- orthogonal to the edge. Here, α_{LF} is the Lax-Friedrichs stabilizer factor. We have two different ways to compute α_{LF} depending on the knowledge of solution.

- The global Lax-Friedrichs Stabilizer: $\alpha_{LF} = \max_u |\partial(\mathbf{f} \cdot \boldsymbol{\nu})/\partial u|$;
- The local Lax-Friedrichs Stabilizer: $\alpha_{LF} = \max(|\partial(\mathbf{f}^+ \cdot \boldsymbol{\nu})/\partial u^+|, |\partial(\mathbf{f}^- \cdot \boldsymbol{\nu})/\partial u^-|)$.

The numerical flux $\mathbf{F}$ (4) is consistent and conservtive across the edge. We now replace the u^+ and u^- with reconstructed values and rewrite the numerical flux as

$$\mathbf{F}(\mathcal{R}^+,\mathcal{R}-) = \frac{1}{2}[(\mathbf{f}(\mathcal{R}^+) + \mathbf{f}(\mathcal{R}^-) \cdot \boldsymbol{\nu} - \alpha_{LF}(\mathcal{R}^+ - \mathcal{R}^-)], \tag{5}$$

where $\mathcal{R}$ is a higher order reconstruction value. We evaluate the integrals over each edge $e_i \in \partial E$ with a quadrature rule. We represent length of edge e_i and area of cell E with $|e_i|$ and $|E|$ respectively. Assuming a quadrature rule with G points on each edge, we have

$$\overline{u}_t + \sum_{i=0}^{4} \frac{|e_i|}{|E|} \sum_{g=0}^{G} \omega_{i,g} \mathbf{F}(\mathcal{R}^+_{i,g}, \mathcal{R}^-_{i,g}) \cdot \boldsymbol{\nu}_i = 0, \tag{6}$$

where $\omega_{i,g}$ represents the weight associated to the g^{th} quadrature points on edge e_i.

0.2 The Multi-level Weighted Essentially Non-Oscillatory (ML-WENO) Method

We develop an accurate, versatile, robust and efficient finite volume weighted essentially non oscillatory (WENO) techniques for reconstruction of discontinuous values with cell averaged values in Arbogast et al. (2024). The ML-WENO reconstruction will be primarily used in reconstruction of values on both sided of the edge for numerical flux computation appears in FVM discretization. It can also be applied to general reconstruction of discontinuous variables, e.g., piece-wise constant discontinuous Darcy pressure appears on the zero-nonzero porosity interface.

0.2.1 Stencil Polynomial

Given a rectangular stencil $\mathcal{S} = \{E_{i,j},\ i \le s,\ j \le t\} \subset \mathcal{T}_h$ of the quasi uniform logically rectangular mesh, which covers the area $\Omega_{\mathcal{S}} = \cup_{E\in\mathcal{S}}\Omega_E$. For each stencil, we associate a tensor product stencil polynomial $Q^{\mathcal{S}}_{s,t} \in \mathbb{Q}_{s-1,t-1}(\Omega_{\mathcal{S}})$ with it. We write the tensor product stencil polynomial with proper scale as

$$Q^{\mathcal{S}}_{s,t}(x,y) = \sum_{p<s,q<t} c_{p,q}\Big(\frac{x-x_{\mathcal{S}}}{h_{\mathcal{S}}}\Big)^p\Big(\frac{y-y_{\mathcal{S}}}{h_{\mathcal{S}}}\Big)^q, \tag{7}$$

where $(x_{\mathcal{S}}, y_{\mathcal{S}})$ is any point in the stencil $\mathcal{S}$. We choose the stencil "center" point: the intersection point of two diagonal lines joining top left, bottom right and top right, bottom left corners. Assuming the cell $E_{i,j} \in \mathcal{S} \subset \mathcal{T}_h$ is shape regular as in Definition **??**, we define the stencil scale parameter $h_{\mathcal{S}} = \sum_{i\le s, j\le t} E_{i,j}/(s\times t)$ as the average diameter of all stencil cells. Let $\xi = (x - x_{\mathcal{S}})/h_{\mathcal{S}}$ and $\eta = (y - y_{\mathcal{S}})/h_{\mathcal{S}}$ denote the normalized variables, we can rewrite equation (7) as

$$\hat{Q}^{\mathcal{S}}_{s,t}(\xi,\eta) = \sum_{p<s,q<t} c_{p,q}\xi^p\eta^q. \tag{8}$$

The tensor product stencil polynomial satisfies the condition that the cell-averaged value (3) is preserved through integral over each cell as

$$\frac{1}{|E_{i,j}|}\iint_{E_{i,j}} Q^{\mathcal{S}}_{s,t}(x,y)\,dxdy = \overline{u}_{i,j}, \quad \forall E_{i,j} \in \mathcal{S}, \tag{9}$$

where $\overline{u}_{i,j}$ represents the cell-averaged scalar variable over cell $E_{i,j}$.

One way to construct a stencil polynomial $Q^{\mathcal{S}}_{s,t}$ is using basis polynomials. Let $Q^{\mathcal{S}}_{s,t} = \sum_{i,j}\overline{u}_{i,j}\phi^{i,j}_{s,t}$, where basis polynomials $\phi^{i,j}_{s,t}$ satisfy

$$\frac{1}{|E_{i',j'}|}\iint_{E_{i',j'}} \phi^{i,j}_{s,t}(x,y)\,dxdy = \begin{cases} 1 & (i',j') = (i,j) \\ 0 & (i',j') \ne (i,j) \end{cases}, \quad \forall E_{i',j'} \in \mathcal{S}, \tag{10}$$

which would always lead to a $(s \times t) \times (s \times t)$ non-singular linear system. We can write the basis polynomials in terms of normalized coordinates ξ and η as well, and represent the normalized stencil polynomial (8) with normalized basis polynomials as:

$$\hat{Q}^{\mathcal{S}}_{s,t}(\xi, \eta) = \sum_{i,j} \overline{u}_k \hat{\phi}^{i,j}_{s,t}(\xi, \eta). \tag{11}$$

In most of the two-dimensional situation, we implement a square stencil, i.e., $s = t = r$. The approxmatibility of tensor product polynomials is given in Dupont and Scott (1980). Given a bounded interpolation projection, when $s \neq t$, the excessive order will be "wasted", since the order of accuracy will be $\min(s, t)$. However, in some cases, where we acquire the prior knowledge of the solution being quasi-1D, e.g., the transport velocity is fixed in one direction. We can have uneven stencils that has $\max(s, t)$ order of accuracy but in the desired direction. In other cases when derivative is involved, we also need an stretched stencil to compensate for the order loss when approximates derivative.

0.2.2 Stencil polynomial smoothness indicators

We define a smoothness indicator σ as a measurement of smoothness of $u(\mathbf{x})$ on the stencil S. The classic method from Jiang and Shu (1996) is defined in analogue to a cell-averaged Sobolev seminorm as:

$$\sigma_{JS} = \sum_{1 \leq |\alpha| \leq m} \int \int_{E_0} |E_0|^{|\alpha|-1} (\mathcal{D}^\alpha Q(x, y))^2 dx dy, \tag{12}$$

where $\alpha = (\alpha_1, \alpha_2)$ is a multi-index, $|\alpha| = \alpha_1 + \alpha_2$, and $\mathcal{D}$ is the derivative operator $\mathcal{D}^\alpha = \partial^{|\alpha|}/\partial x^{\alpha_1} \partial y^{\alpha_2}$. The Jiang-Shu smoothness indicator can be reformed with basis polynomials:

$$\begin{aligned}
\sigma_{JS} &= \sum_{1 \leq |\alpha| \leq m} |E_0|^{|\alpha|-1} \iint_{E_0} (\mathcal{D}^\alpha Q(x, y))^2 dx dy \\
&= \sum_{1 \leq |\alpha| \leq m} \iint_{\hat{E}_0} (\mathcal{D}^\alpha \hat{Q}(\xi, \eta))^2 d\xi d\eta \\
&= \sum_{1 \leq |\alpha| \leq m} \iint_{\hat{E}_0} \big(\sum_{ij} \overline{u}_{ij} \mathcal{D}^\alpha \hat{\phi}_{i,j}(\xi, \eta) \big)^2 d\xi d\eta \\
&= \sum_{ij} \sum_{i'j'} \overline{u}_{ij} \overline{u}_{i'.j'} \sum_{1 \leq |\alpha| \leq m} \iint_{\hat{E}_{ij}} \mathcal{D}^\alpha \hat{\phi}_{ij}(\xi, \eta) \mathcal{D}^\alpha \hat{\phi}_{i'j'}(\xi, \eta) d\xi d\eta \\
&= \sum_{ij} \sum_{i'j'} \overline{u}_{ij} \overline{u}_{i'j'} \sigma^{i'j'}_{ij},
\end{aligned} \tag{13}$$

where $\sigma_{ij}^{i'j'}$ can be precomputed after mesh is established after mesh is established.

Alternatively, we can expand the Jiang-Shu smoothness indicator according to polynomial coefficients, which paves the way for further simplification. Recall the tensor product polynomial version (8),

$$\begin{aligned} \sigma_{JS} &= \sum_{1\leq|\alpha|m} \\ &= \sum\sum \eta_i c_i c_j \end{aligned} \tag{14}$$

We observe that the smoothness indicator exhibit following convergence property that there exists some value $C > 0$ such that when $h \to 0^+$,

$$\sigma = \begin{cases} Ch^2 + \mathcal{O}(h^3) & \text{Smooth,} \\ \Theta(1) & \text{Not Smooth,} \end{cases} \tag{15}$$

where stencils are defined on quasi uniform mesh. This observation comes directly from the scaling property of Sobolev seminorm.

0.2.3 Two alternative smoothness indicator

We proposed two alternative ways of defining smoothness indicator in Huang et al. (2023) and Arbogast et al. (2025).

- *Simple smoothness indicator*

$$\sigma_{\text{simp}} = \frac{1}{N_{\mathcal{S}} - 1} \sum_k (\overline{u}_k - \overline{u}_0)^2, \tag{16}$$

  where $\overline{u}_0$ is the cell averaged solution in target cell. We define this simple smoothness indicator in order to reduce the computational cost and to deal with distorted stencil on unstructured mesh. However, this simple smoothness indicator is not as robust as the classic Jiang-Shu smoothness indicator. Hence, we stick to the classic definition in the following discussion.

- *Polynomial smoothness indicator*

$$\sigma_P = \sum \eta_i c_i^2 \tag{17}$$

  We obtain this polynomial smoothness indicator by dropping all the cross terms.

0.2.4 ML-WENO weighting

Given a target cell $E_{i,j}$, we can create a nonempty set of stencils $\{\mathcal{S}_l \ni E_{i,j}, l = 0, 1, \dots, L, L \geq 0\}$. The reconstruction of u on target cell $E_{i,j}$ is a convex combination of stencil polynomials (11)

$$\mathcal{R}(\mathbf{x}) = \sum_l \tilde{\omega}_l Q_l(\mathbf{x}), \quad \mathbf{x} \in E_{i,j}, \tag{18}$$

where $\sum_l \tilde{\omega}_l = 1$ is normalized nonlinear weights associated with stencils accordingly. We begin by arbitrarily chosing a set of linear weights $\{\omega_l\}$ and scale with stencil

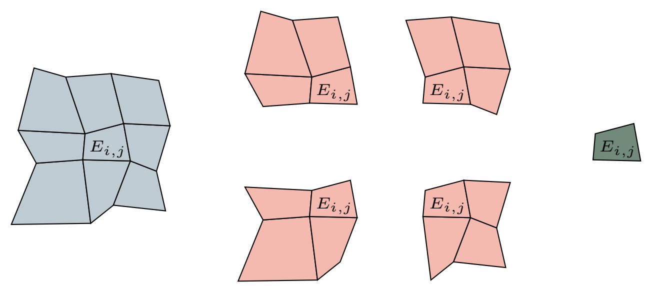

Figure 1: An example of (3,2,1) multi-level stencils combination on logically rectangular quadrilateral mesh.

smoothness indicator as

$$\hat{\omega}_l = \frac{\omega_l}{(\sigma_l + \epsilon h_0^2)^{ar_l + n_l}}, \tag{19}$$

where $r_l = \max\{s, t\}$, $\epsilon = 1e - 4$ and $a > 0.5$ are picked as constants. The biasing factor $n_l = n(r_l)$ is used to biasing away further from constant level, whose smoothness indicator is always zero, when the solution is smooth. We pick

$$n(r_l) = n_l = \begin{cases} 1, & \text{if} \quad r_l = 1, \\ 3, & \text{if} \quad r_l = 2, \\ 4, & \text{if} \quad r_l \geq 3. \end{cases} \tag{20}$$

We then renormalize scaled weights

$$\tilde{\omega}_l = \frac{\hat{\omega}_l}{\sum_{l'} \hat{\omega}_{l'}}. \tag{21}$$

0.2.5 The ML-WENO Reconstruction Error

In this section, we consider the accuracy of our multi-level weighting scheme. We group the set of indices $l \in \mathcal{L} = \{0, 1, 2, \dots L\}$ according to the order of reconstruction polynomials, i.e., the shape of individual stencil. We denote the set of indices of stencils have s cells in x direction and t cells in y direction as $\mathcal{L}_{s,t}$. We make the assumption that the mesh resolution is fine enough so that there exist at least one stencil that is smooth if shock presents. We define two sets holding indices of stencils with or without jump as

$$\begin{aligned} \text{Smooth} &= \{l : \ u \text{ is smooth on } \mathcal{S}_l, \} \\ \text{Jump} &= \{l : \ u \text{ has a jump on } \mathcal{S}_l\} \end{aligned} \tag{22}$$

where we assume Smooth $\cup$ Jump $= \mathcal{L}$ and Smooth $\cap$ Jump $= \emptyset$. We represent the set of indices of stencils of shape (s,t) that is smooth as $\mathcal{L}_{s,t} \cap$ Smooth, and the set of stencils of shat (s,t) that has a jump as $\mathcal{L}_{s,t} \cap$ Jump. For convience, we assume square stencils, where $r = s = t$. The accuracy result can be applied to $s \neq t$ situation directly in the selected direction. We show that the reconstruction is accurate with respect to the target cell E_0 to order

$$r_{\max} = \max\{r : \text{Smooth} \cap \mathcal{L}_{r,r} \neq \emptyset\}. \tag{23}$$

In Figure 2, we illustrate a (3,2,1) combination of stencils with thow shock presents. Here, Smooth $= \{0, 1\}$, which indicates that stencil $\mathcal{S}_0$ and $\mathcal{S}_1$ do not have jump present. In this case, we expect the accuracy of reconstruction is $r_{\max} = \text{Smooth} \cap \mathcal{L}_{2,2} = 2$.

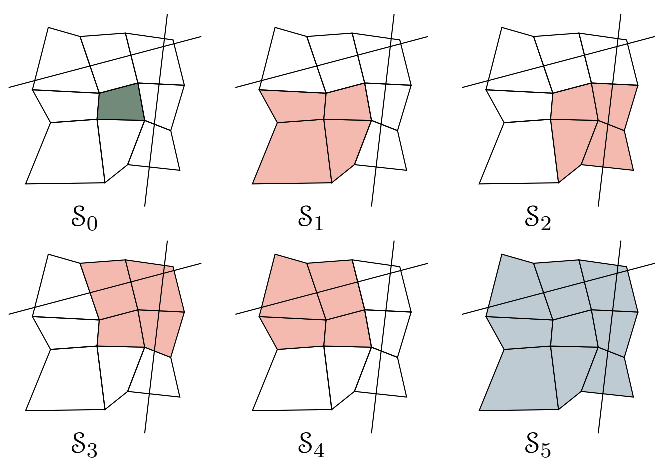

Figure 2: An illustration of a (3,2,1) multi-level stencils combination on logically rectangular quadrilateral mesh with two shock presents.

We state the estimation of reconstruction error show in Arbogast et al. (2024). We start by estimating the error of our nonlinear scaling (19) asscoiated to some stencil $\mathcal{S}_l$ for some $m > 0$.

$$\hat{\omega}_l = \frac{\omega_l}{(\sigma_l + \epsilon h_0^2)^m}. \tag{24}$$

We estimate the reconstruction error for two situtaions.

- $l \in$ Smooth, the smoothness indicator $\sigma_l = Ch^2 + \mathcal{O}(h^3)$;
- $l \in$ Jump, the smoothness indicator $\sigma_l = \Theta(1)$.

Use the asymptotic estimation result from Olver (1997). Let $n \geq m$, $a > 0$, and $b > 0$

$$\begin{aligned} \frac{a}{bh^m + \mathcal{O}(h^n)} &= \frac{ah^{-m}}{b}\big(1 + \mathcal{O}(h^{n-m})\big), \\ \frac{a}{bh^{-n} + \mathcal{O}(h^{-m})} &= \frac{ah^n}{b}\big(1 + \mathcal{O}(h^{n-m})\big). \end{aligned} \tag{25}$$

For the smooth stencil $\mathcal{S}_l \in$ Smooth

$$\begin{aligned} \hat{\omega}_l = \frac{\omega_l}{(\sigma_l + \epsilon h_0^2)^m} &= \frac{\omega_l}{((C+\epsilon)h_0^2 + \mathcal{O}(h^3))^m} \\ &= \omega_l \Big(\frac{h_0^{-2}}{C+\epsilon}(1 + \mathcal{O}(h))\Big)^m \\ &= \frac{\omega_l}{(D+\epsilon)^m} h_0^{-2m}(1 + \mathcal{O}(h)). \end{aligned} \tag{26}$$

In conclusion, we estimate the nonlinear scaling as

$$\hat{\omega}_l = \begin{cases} \dfrac{\omega_l}{(C+\epsilon)^m} h_0^{-2m}(1 + \mathcal{O}(h)), & \text{if} \quad l \in \text{Smooth}, \\ \Theta(1), & \text{if} \quad l \in \text{Jump}. \end{cases} \tag{27}$$

Then we estimate the error for the normalized nonlinear weighting (21). We expand the sum of nonlinear scaling as

$$\sum_l \hat{\omega}_l = \sum_r \sum_{\substack{l \in \\ \text{Smooth} \cap \mathcal{L}_{r,r}}} \hat{\omega}_l + \sum_{l \in \text{Jump}} \hat{\omega}_l. \tag{28}$$

We estimate the smooth part of the nonlinear weighting as

$$\sum_r \sum_{\substack{l \in \\ \text{Smooth} \cap \mathcal{L}_{r,r}}} \hat{\omega}_l = \sum_r \sum_{\substack{l \in \\ \text{Smooth} \cap \mathcal{L}_{r,r}}} \frac{\omega_l}{(C+\epsilon)^m} h_0^{-2(ar+\eta)}(1 + \mathcal{O}(h)). \tag{29}$$

Assume $h<1$, we keep the largest $ar+\eta(r)$ number as our estimator

$$\begin{aligned}\sum_r \sum_{\substack{l\in\\ \text{Smooth}\cap\mathcal{L}_{r,r}}} \hat{\omega}_l =& \sum_{\substack{l\in\\ \text{Smooth}\cap\mathcal{L}_{r_{\max},r_{\max}}}} \frac{\omega_l}{(C+\epsilon)^{ar_{\max}+\eta(r_{\max})}} h_0^{-2(ar_{\max}+\eta(r_{\max}))}(1+\mathcal{O}(h)) \\ &+\Theta(h^{-2(ar+\eta(r))}) \\ =& \sum_{\substack{l\in\\ \text{Smooth}\cap\mathcal{L}_{r_{\max},r_{\max}}}} \frac{\omega_l}{(C+\epsilon)^{ar_{\max}+\eta(r_{\max})}} h_0^{-2(ar_{\max}+\eta(r_{\max}))} \\ &+\mathcal{O}(h^{1-2(ar_{\max}+\eta(r_{\max}))}).\end{aligned} \tag{30}$$

later, we denote the set $\text{Smooth}\cap\mathcal{L}_r$ as Smooth_r for simplicity. The overall accuracy estimation of the nonlinear weighting (21) is presented as

- $l\in\text{Smooth}$
$$\tilde{\omega}_l = \frac{\hat{\omega}_l}{\sum_{l'}\hat{\omega}_{l'}} = \frac{\omega_l}{\sum_{l'\in\text{Smooth}_r}\omega_{l'}} h^{2[a(r_{\max}-r)+\eta(r_{\max})-\eta(r)]}(1+\mathcal{O}(h)); \tag{31}$$
- $l\in\text{Jump}$
$$\tilde{\omega}_l = \frac{\hat{\omega}_l}{\sum_{l'}\hat{\omega}_{l'}} = \Theta(h^{2[a(r_{\max})+\eta(r_{\max})]}). \tag{32}$$

Now we can estimate the reconstruction error as

$$\begin{aligned}|u(\mathbf{x})-\mathcal{R}(\mathbf{x})| &= |\sum_l \tilde{\omega}_l(u(\mathbf{x}-Q_l(\mathbf{x})))| \\ &\le \sum_{l\in\text{Smooth}} \tilde{\omega}_l|u(\mathbf{x})-Q_l(\mathbf{x})| + \sum_{l\in\text{Jump}} \tilde{\omega}_l|u(\mathbf{x})-Q_l(\mathbf{x})| \\ &\le \sum_r \sum_{\substack{l\in\\ \text{Smooth}\cap\mathcal{L}_{r,r}}} \Theta(h^{2[a(r_{\max}-r)+(\eta(r_{\max})-\eta(r))]})\mathcal{O}(h^r) \\ &\quad+\sum_{l\in\text{Jump}}\Theta(h^{2(ar_{\max}+\eta(r_{\max}))})\mathcal{O}(1) \\ &= \mathcal{O}(h^{r_{\max}})\end{aligned} \tag{33}$$

Combining everything above and (18), we arrive at the formal error estimate of the reconstruction.

Lemma 0.1. *Let $\mathcal{R}(\mathbf{x})$ be the reconstruction. Assuming a qusi-uniform mesh, $h_0 = \Theta(h)$, there exists a constant $C>0$ such that for any $\mathbf{x}\in E_0$,*

$$|u(\mathbf{x})-\mathcal{R}(\mathbf{x})| \le Ch^{r_{\max}}, \tag{34}$$

where $r_{\max}$ is defined by (23).

0.2.6 Handling Boundary Conditions

In our simulation, we couple FEM and FVM solver on a single mesh. Therefore, accurate and compatible boundary condition should be assigned accordingly. However, it is usually difficult to manage boundary condition for higher order finite volume and finite difference method, which require wide stencils. Some deal with the difficulty use ghost cells or points outside the boundary, and maintain the same stencil as if they were in the interior of the domain. However, the extrapolation or estimation of the manufactured values of ghost cells or ghost points are not straightforward. A notable method, the inverse Lax-Wendroff procedure, substitute the normal derivative of the solution with tangential and time derivatives using the PDE relation in an iteratvie way to impose more precise values for the ghost cells or points. The key to the above strategy is to have accurate and stable reconstruction near the boundary. ML-WENO gives the flexibility to add stenncils of the desired size that span back into the domain.

In order to assign proper boundary condition to a FVM discretization, we should determine whether the flow is inflow or outflow first. If $\mathbf{f}(u_B) \cdot \boldsymbol{\nu} > 0$, we recognize that it is a outflow boundary, therefore, no fixed boundary condition should be assigned. We use "free outflow" boundary that the numerical flux is simply $\mathbf{f}(u_B) \cdot \boldsymbol{\nu}$.

The inflow boundary can be understood in various ways. We can understand it as prescribing the inflow flux $\mathbf{f}_B$ or fixing the boundary value u_B as an equivalent to Dirichlet boundary in the context of Finite Element. In the special case when a zero flux is prescribed, we would call it noflow boundary. In the inflow case, there may be a discontinuous inflow shock wave. We require a swift response inside the domain to align with the specified Dirichlet value. We therefore add an extra constant stencil polynomial to the inflow boundary to accommodate the potential discontinuity.

There are some other special cases that require special treatment on the boundary, e.g., when symmetry should be enforced.

0.2.7 Numerical Test

For completeness, we test our ML-WENO algorithm with simple Riemann shock and rarefraction. Let $\mathbf{f}(u) = (u^2/2, u^2/2)$ be the Burgers flux function. In this example, we take the initial condition $u_0(x, y) = 1$, and the computational domain $\Omega = [0, 3] \times [0, 1]$. The true solution is

$$u(x,y,t) = \begin{cases} 0, & x < 0.5, x \geq 0.5t + 1.5, \\ (x-0.5)/t, & 0.5 \leq x < t + 0.5, \\ 1, & t + 0.5 \leq x < 0.5t + 1.5. \end{cases} \tag{35}$$

We forward marching to the time = 2.0, when trailing rarefaction reaches the leading shock. We use SSP2RK time stepping scheme.

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